

2-6

convert dec. to hex as IEEE 754: $(-1)^s \cdot 1.F \cdot 2^{E-127}$

a) 1.0 $\rightarrow$ positive, $s=0$

$$\text{num} = \text{mantissa} \cdot 2^{\text{exponent}} \rightarrow 1.0 = 1 \cdot 2^0 = 1 \cdot 0^0 = \underline{1.0}$$

$$\text{exponent} = 0 + 127 = 127$$

$$\hookrightarrow 2^6 + 2^5 + 2^4 + 2^3 + 2^2 + 2^1 + 2^0$$

$$= \underline{01111111}$$

$$s=0 \quad E = 01111111 \quad F = 000000000000000000000000$$

$$\hookrightarrow \text{bc } 1.0 \text{ has no frac}$$

$$= \underline{00111111} \underline{000000000000000000000000}$$

$$= 3F800000 = \underline{0 \times 3F800000}$$

b) -16.5 $\rightarrow$ negative, $s=1$

$$16 = 2^4, \quad 0.5 = 2^{-1}$$

$$\begin{array}{c} 11 \\ 10100 \end{array}$$

$$\overline{11}$$

$$0.1 \Rightarrow 16.5 = 1000.1 \cdot 2^4 = \underline{100001}$$

$$E = 4 + 127 = 131$$

$$131/2 = 65 \text{ R } 1, 65/2 = 32 \text{ R } 1, 32/2 = 16 \text{ R } 0$$

$$16/2 = 8 \text{ R } 0, 8/2 = 4 \text{ R } 0, 4/2 = 2 \text{ R } 0, 2/2 = 1 \text{ R } 0$$

$$1/2 = 0 \text{ R } 1$$

$$131 = 11000001 = \underline{100000011}$$

$$\text{mantissa: } 1.00001 = 000010000000000000000000$$

$$\underline{11000001} \underline{100001} \underline{000000000000000000000000}$$

$$C1840000$$

$$= \underline{0 \times C1840000}$$

c) $314.25 \rightarrow$ pos, s = 0 : binary: $314/2 = 157, 157/2 = 78 \text{ r } 1, 78/2 = 39$
 $39/2 = 19 \text{ r } 1, 19/2 = 9 \text{ r } 1, 9/2 = 4 \text{ r } 1, 4/2 = 2, 2/2 = 1, 1/2 = 1 \text{ r } 1$
 $= 100111010:$

$$0.25 \cdot 2 = .5, .5 \cdot 2 = 1, 0.25 = 0.01$$

$$\underline{314.25 = 100111010.01}$$

$$1.F \times 2^E = 100111010.01 \cdot 2^8 = \underline{1.0011101001}$$

$$\text{mantissa} = 0011101001$$

$$\text{exponent} = 8 + 127 = 135 \rightarrow 10000111$$

all together 0100001110011010100000000000
439D2000